

Siavash Shami
M.Sc. in Geodesy | InSAR & Remote Sensing Researcher
Faculty of Geodesy and Geomatics Engineering
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About

Siavash Shami is a geodesist and InSAR researcher specializing in SAR-based surface deformation monitoring and Earth observation. His research focuses on InSAR methodology, interferometric time-series analysis, error characterization, and the integration of InSAR with GNSS, remote sensing, geological, and hydrological data to investigate natural and anthropogenic deformation. He also applies machine learning and deep learning to deformation modeling and environmental remote sensing.

Research Interests

- InSAR and SAR Remote Sensing
- Surface and Crustal Deformation Monitoring
- InSAR Time-Series Analysis, Network Design, and Error Characterization
- Geohazards and Natural & Anthropogenic Deformation
- Remote Sensing for Environmental and Climate Change Studies
- Machine Learning and Deep Learning for Earth Science Applications

Education

- **M.Sc. in Geodesy** - K. N. Toosi University of Technology, Iran | 2018
Thesis: *A Feasibility Study on the Application of InSAR for Surveying in Open Pit Mines* [In Persian]
- **B.Sc. in Surveying Engineering** - Eyn Ol Ghozat Higher Education Institute, Iran | 2016

Experiences

Scientific Reviewer

- Gondwana Research - Elsevier (1)
- Earth Science Informatics - Springer Nature (1)
- Geography, environment, sustainability (1)
- IEEE Access - IEEE (1)
- Advances in Space Research - Elsevier (1)
- Remote Sensing Applications: Society and Environment - Elsevier (3)
- Revue Internationale de Géomatique - Tech Science Press (1)

Awards and Honors

- Ranked 1st among B.Sc. students of Surveying Engineering (GPA: 3.93/4.00) | 2016

Research & Professional Experiences

- Research Collaboration - InSAR Applications | 2023–2024
[Collaborated with Dr. Meisam Amani on InSAR-based investigations of landslides in New Zealand, dam deformation and shoreline changes in Canada, & open-pit mine deformation in South Africa]
- InSAR & GIS Consultant/Research Collaborator - Bonyan Zamin Co. | 2022
[Prepared annual InSAR-based surface displacement maps of Isfahan, Iran, for Isfahan Municipality, using Sentinel-1 data, PS-InSAR, GMTSAR, StaMPS, and QGIS]
- Military Service | 2019-2021
[Served as a full-time conscript; during this period, he continued independent research and technical writing and authored a practical handbook on InSAR processing]

Skills

InSAR & SAR Remote Sensing

- SAR Data: Sentinel-1, ENVISAT, ALOS-1/2, ERS-1/2
- InSAR Methods: PS-InSAR, SBAS, NSBAS, DInSAR
- InSAR Processing: Multilooking, noise reduction, interferogram network design and optimization
- Error Characterization & Corrections: Atmospheric correction (ERA5, GACOS), filtering, phase-bias analysis
- Software & Processing Tools: StaMPS/MTI, GMTSAR, MintPy, ISCE, LiCSBAS, SNAP, DORIS, ROI_PAC

Geodesy & Geodetic Techniques

- Satellite Geodesy and Physical Geodesy
- GNSS: Dual-Frequency GNSS
- Levelling and Topographic Surveying
- Total Station and Nivo

Remote Sensing & Environmental Applications

- Applications: Soil Moisture, Groundwater Storage, Precipitation, Land Cover, Air Pollution, Climate change, Soil Erosion
- Data & Products: Landsat, Sentinel-1/2, MODIS, GRACE, Well Data, CHIRPS, TRMM, Sentinel-5P/TROPOMI, MODIS

AI & Data Analysis

- Machine Learning: Random Forest, XGBoost, SVM, Linear Regression, Lasso Regression
- Deep Learning: RNN, LSTM

Programming, GIS & Software

- Programming & Data Analysis: Python, NumPy, Pandas, Matplotlib
- GIS & Remote Sensing Platforms: QGIS, Google Earth Engine
- Surveying & Geospatial Software: Civil 3D, Global Mapper, RTKLIB
- Operating Systems: Linux (Ubuntu)

Languages

- Azerbaijani: Native
- Persian: Native
- English: Professional Working Proficiency

Publications

17. Karami, E., **Shami, S.**, Maghsoudi, Y., Ranjgar, B., & Azadnejad, S. (2025). Investigating the InSAR Phase Bias in the SBAS Algorithm and Its effect on different Landcovers. *IEEE Access* (peer-reviewed: Research Article)
doi.org/10.1109/ACCESS.2025.3568144
16. Naboureh, A., Li, A., Bian, J., Lei, G., Nan, X., Zhang, Z., **Shami, S.** & Lin, X. (2024). Green space coverage versus air pollution: a cloud-based remote sensing data analysis in Sichuan, Western China. *International Journal of Digital Earth* (peer-reviewed: Research Article)
doi.org/10.1080/17538947.2024.2383454
15. **Shami, S.**, Shahriari, M. A., Nilfouroushan, F., Forghani, N., Salimi, M., & Reshadi, M. A. M. (2024). Surface displacement measurement and modeling of the Shah-Gheyb salt dome in southern Iran using InSAR and machine learning techniques. *International Journal of Applied Earth Observation and Geoinformation* (peer-reviewed: Research Article)
doi.org/10.1016/j.jag.2024.104016
14. Abdalla, A., **Shami, S.**, Shahriari, M. A., & Azar, M. K. (2024). Estimation of land displacement in East Baton Rouge Parish, Louisiana, using InSAR: Comparisons with GNSS and machine learning models. *The Egyptian Journal of Remote Sensing and Space Sciences* (peer-reviewed: Research Article)
doi.org/10.1016/j.ejrs.2024.02.008
Co-first author (equal contribution)
13. Khoshlahjeh Azar, M., **Shami, S.**, Nilfouroushan, F., Salimi, M., Ghayoor Bolorfroshan, M., & Reshadi, M. A. M. (2022). Integrated analysis of Hashtgerd plain deformation, using Sentinel-1 SAR, geological and hydrological data. *Scientific Reports* (peer-reviewed: Research Article)

doi.org/10.1038/s41598-022-25659-4
Co-first author (equal contribution)

12. **Shami, S.**, Azar, M. K., Nilfouroushan, F., Salimi, M., & Reshadi, M. A. M. (2022). Assessments of ground subsidence along the railway in the Kashan plain, Iran, using Sentinel-1 data and NSBAS algorithm. *International Journal of Applied Earth Observation and Geoinformation* (peer-reviewed: Research Article)
doi.org/10.1016/j.jag.2022.102898

11. **Shami, S.**, Ranjgar, B., Bian, J., Khoshlahjeh Azar, M., Moghimi, A., Amani, M., & Naboureh, A. (2022). Trends of CO and NO2 Pollutants in Iran during COVID-19 pandemic using Timeseries Sentinel-5 images in Google Earth Engine. *Pollutants* (peer-reviewed: Research Article)
doi.org/10.3390/pollutants2020012

10. **Shami, S.** (2021). GMTSAR installation and processing Guide - Practical InSAR handbook series. *ebook* (Book: Handbook)
doi.org/10.5281/zenodo.22609191

9. **Shami, S.**, Khoshlahjeh, M., Ghorbani, Z., Moghimi, A., Mohammadzadeh, A., & Sabet Ghadam, S. S. (2021). Evaluation of air pollution contributes for the COVID-19 pandemic in Iran using Sentinel 5 Satellite Data. *Journal of Geomatics Science and Technology* (In Persian) (peer-reviewed: Research Article)

8. Hossainali, MM., & **Shami, S.** (2020). Processing Radar Images using StaMPS. *K.N. TOOSI UNIVERSITY Press* (In Persian, ISBN: 978-622-6655-46-0) (Book: Technical Book)
Co-first author (equal contribution)

7. **Shami, S.**, & Ghorbani, Z. (2019). Processing Radar Images using GMTSAR & SNAP. *Arshadan Press* (In Persian, ISBN: 978-622-251-021-3) (Book: Technical Book)

6. **Shami, S.**, & Ghorbani, Z. (2019). Investigating water storage changes in Iran using grace and chirps data in the google earth engine system. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* (Conference Paper: Poster Presentation)
doi.org/10.5194/isprs-archives-XLII-4-W18-981-2019

5. **Shami, S.**, & Ghorbani, Z. (2019). Influence of wavelength radar images in estimation of open-pit mine displacements using radar interferometry technology (case study: Gol Gohar Sirjan mine). *3rd TRIGGER International Conference* (Conference Paper: Poster Presentation)

4. **Shami, S.**, Hossainali, MM., & Babaee, S. (2019). Analysis of Large-scale Displacement Using Radar Interferometry Technology in Open-pit Mines (Case Study: Gol Gohar Sirjan Mine). *Geospatial Engineering Journal* (In Persian) (peer-reviewed: Review Article)

3. **Shami, S.**, Ghorbani, Z., & Abedi, L. (2018). Calculation of the extracted mines volume using DEM obtained from the Envisat radar satellite images (Case study: Gol Gohar Sirjan Mine). *The 4th National Geology and Mining Explorations Symposium* (In Persian) (Conference Paper: Oral Presentation)

2. **Shami, S.**, Ghorbani, Z., & Abedi, L. (2018). Persistent Scatterers Time series analysis of radar interferometry to determine the rate subsidence of areas around the open pit mines (Case study: Gol Gohar Sirjan Mine). *The 4th National Geology and Mining Explorations Symposium* (In Persian) (Conference Paper: Poster Presentation)

1. **Shami, S.**, & Hossainali, MM. (2018). Analysis of the effect of temporal and spatial baseline of radar images on the amount of coherence for detecting large-scale displacement gradients using interferometric synthetic aperture radar (Case study: Gol Gohar Sirjan Mine). *Geomatics and GIT 97* (In Persian) (Conference Paper: Poster Presentation)